ENGR 3100

The Problem of E-Waste

A look at the current problem, and suggestions for the future.

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5-4-2026

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Abstract

Electronic waste (E-waste) is an ever-growing global issue. The facilities to handle the rate at which electronics enter the market and, in turn, enter our waste streams do not exist. Thus, as electronics continue to enter and exit the market, they are disposed of improperly. Electronics often have precious metals like gold and silver that then get wasted and heavy metals like lead and mercury that seep into the environment. As more e-waste is generated, the economic waste and environmental harm will only grow alongside it if the proper measures are not put into place. In the United States, there is no federal legislation regarding e-waste, and while some states have legislation, it is inconsistent between states leading to confusion within the U.S. recycling industry. This paper looks at the current issue of e-waste and proposes economic, manufacturing, and legislative solutions to mitigate the problem.

Introduction

What is E-Waste?

Electronic waste (e-waste) can be defined in several different manners, the simplest of which being the waste or disposal of any product requiring electricity. However, there does not exist a consensus on what defines e-waste. The U.S. Environmental Protection Agency (EPA), defines e-waste as a subset of electronics pertaining to video, audio, and information products [1], [2]. While in the *Global E-Waste Monitor* by the United Nations Institute for Training and Research (UNITAR), e-waste is defined via electrical and electronic equipment (EEE) definition of e-waste, which is any product "with circuitry or electrical components and a power or battery supply" [3]. UNITAR's *Global E-Waste Monitor* also sorts e-waste into six subcategories as seen in table 1. The discrepancies in defining e-waste are important to note as it greatly affects how e-waste is monitored and measured. A narrower definition allows for the negligence of specific categories of generated waste and makes policing e-waste a harder task. For the sake of this paper, UNITAR's definition of e-waste will be what is referred to when e-waste is mentioned.

E-Waste Category	Examples
Temperature Exchange Equipment	Refrigerators, freezers, air conditioners, and heat pumps
Screens and Monitors	Televisions, monitors, laptops, and tablets
Lamps	Fluorescent lights, high-intensity discharge lamps, and LEDs.
Large Equipment	Washing Machines, clothe dryers, dishwashers, electric stoves, and solar panels.
Small Equipment	Vacuum cleaners, toasters, electric razors, calculators, radios, cameras, and e-cigarettes,
Small IT and Telecommunication Equipment	Phones, personal computers, GPS devices, routers, and printers.

Table 1: *Categorizations of e-waste and examples from each category adopted from [3].*

What Causes E-waste?

"The world is experiencing significant electronification," as UNITAR's 2024 *Global E-waste Monitor* puts it. Electronics enter consumer markets every year at an increasing rate [3]. As these electronics enter the market, the waste generated by them increases as well, as seen in

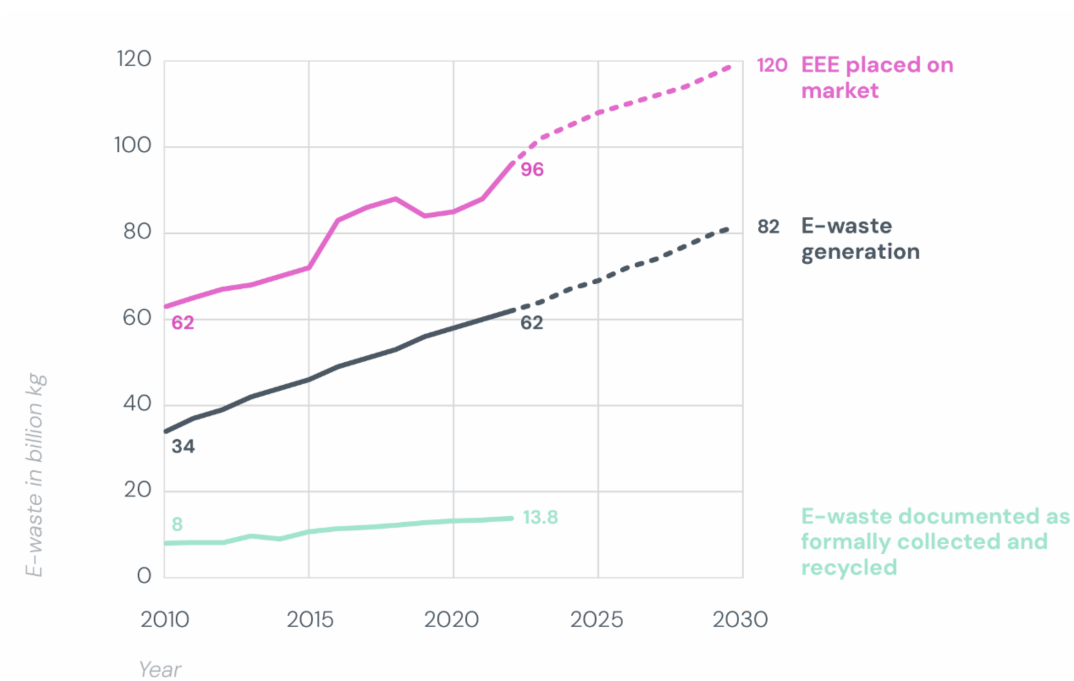

Figure 1: *Plot displaying how the rate of EEE placed on the market grows alongside the generation of e-waste, and how their rates of growth is larger than the formal recycling of e-waste. Taken from [3].*

figure 1. It can also be seen in this figure that the ever-increasing generation of e-waste does not match our abilities to handle said waste. In the Americas, only 30% of the e-waste generated is properly collected, the remaining 70% is sent to improper waste streams [3].

The ever-increasing implementation of electronics into a consumer's life does not necessarily mean that e-waste is going to be generated, and yet we see a direct correlation between the two in figure 1. This is in part due to the concept of planned obsolescence. Planned obsolescence is the designing of a product with an artificially useful lifetime or designing it to be purposefully frail so that it becomes obsolete in a predetermined amount of time [4], [5]. This concept of obsolescence comes in physical and social forms.

Physically, planned obsolescence is when an electronic device is rendered useless either due to some physical impairment or software limiting the usefulness of the device. This is seen when a product is designed with the intention of breaking without the ability to repair it, or a device is no longer supported by a software distributor. An example of the latter was seen in 2024, when music streaming company Spotify fully stopped supporting their music playing device the Car Thing, disabling the supported software and rendering the otherwise functional device useless for people who had purchased it [6].

Socially, as new electronics enter consumer markets, people are marketed to with the intention of throwing out their old devices to get what is newer and better. Proske and Jager-Erben present this idea of "mental deprecation", where perceiving a device as old and outdated is in part due to the age of the product and in part due to "pre-existing expectations regarding the product, experiences with other products (especially regarding the lifetime), and by practical knowledge about upgrades and successors,". Surveys have shown that smartphones are often only used for two years by their first users and still work when replaced [5]. This social need to get a newer device due to a change in look or updated features falls under the obsolescence umbrella and leads to the generation of e-waste.

The goal of this paper is to highlight the growing problem of e-waste. It looks at the current methods of handling e-waste and the economic and environmental impacts that increases in e-waste every year have on the world. From this, solutions are posed that could potentially mitigate these effects. The objective of this paper is not to present one primary solution, but to present many ideas that could tackle e-waste from various angles based on the current problems e-waste poses.

Background

How is E-waste Currently Handled?

As e-waste grows, efforts around the globe are being made to minimize the impact of e-waste. From legislation meant to regulate shipment and disposal of e-waste, to limitations on what materials electronic devices can be comprised of. There are also efforts being made regarding recycling technologies that are meant to break down e-waste and separate the harmful and/or valuable materials out of the disposed devices.

Current Legislation

There exists legislation intended to minimize the impact of e-waste on the environment, economic waste, and to better handle its growth. 81 of 193 countries in the United Nations, and 25 of 50 states in the U.S., plus the District of Columbia, have some amount of legislation regarding e-waste [3], [7].

In terms of global legislation, the most notable one is the Basel Convention, a global treaty regarding the transboundary movement of hazardous waste such as e-waste. This convention was negotiated in the mid-1980s and put into full effect with the Basel Ban Amendment in the mid-1990s, putting greater restrictions on the transportation of hazardous waste to poor countries [1], [8]. However, while 178 countries are party to the Basel Convention and 55 countries have signed and ratified the Basel Ban, the U.S. is the only developed country to have signed but not ratified it. There is also the Restriction of Hazardous Substances directive (RoHS), a directive introduced in 2003 by the EU, to limit the usage of certain heavy metals and common toxic chemicals in electronics [9]. The EU also introduced the Waste from Electrical and Electronic Equipment (WEEE) directive in 2012. A directive meant to fully prevent the creation of e-waste by promoting reuse, recycling, and recovering materials over the disposal of electronics [10]. The Basal convention is the only truly global e-waste legislation, and the EU's RoHS and WEEE have inspired talks of legislation elsewhere, but there is still a significant lack of comprehensive e-waste legislation worldwide.

The U.S. is one of the only developed countries without country-wide legislation on e-waste. However, despite the lack of federal action regarding e-waste in the U.S., half of the states have some state-based legislation regarding e-waste, but they are all different from one another.

This leads to a "'patchwork' of disparate regulations" that creates challenges in the recycling industry due to the differing rules between states [11]. Standard solid waste collection in the U.S., as defined by the Resource Conservation and Recovery Act (RCRA) and the Hazardous and Solid Waste Amendment (HSWA), are not capable of managing e-waste within their scope due to there being numerous exemptions and exclusions in their policies [1], [12].

The RCRA and HSWA are ultimately not equipped to regulate e-waste disposal. E-waste exports to other countries are only illegal under the RCRA if the waste is meant to be disposed of overseas, but if it is meant to be "recycled" U.S. shippers can legally send e-waste anywhere. Additionally, e-waste recyclers within the U.S. can only be regulated by the RCRA if they generate hazardous waste, but if the electronic device is shipped off before being disassembled, it becomes classified as a commodity. E-waste recyclers can ship commodities anywhere without needing to follow the RCRA's recycler requirement of providing the EPA prior notification of shipment and obtaining consent from the nation receiving the waste [12].

Additionally, while 25 states and the District of Columbia have legislation regarding e-waste disposal, they are all independent of one another. 23 states have implemented extended producer responsibility (EPR) policies in their e-waste legislation, requiring manufacturers to pay an annual registration fee and set up an e-waste recycling program in the state. These programs can vary from offering the ability to mail in e-waste products to having dedicated drop off centers in a city [13], [14]. This shifts end-of-life responsibility of consumer electronics to the producer and away from the public sector, incentivizing producers to consider the environment in the design of their products, but the specific manners each state mandates EPR varies greatly. The products covered and metrics of performance are also inconsistent between states. This creates confusing legislation for both producers and recyclers across state boarders, which can lead to no regulations being followed at all [11], [12].

Current Recycling Practices

When e-waste is properly recycled, the process of dismantling devices can be both costly, time-intensive, and dangerous if not done correctly. Printed circuit boards (PCBs), for example, require desoldering the chips from the board, and then open burning and acid baths to remove the metals from the board. This process emits metals like tin, lead, and mercury into the air [15]. With current technologies, manual disassembly of electronics yields a higher gain of recovered metals. A difference of around 600g/metric ton of gold can be recovered from a phone if done by hand rather than by a machine [16]. However, this by hand process can be dangerous for workers without the proper personal protective equipment.

Fortunately, with the growth of e-waste there is also a growth in treatment technologies. From 2010 to 2022, there was an increase in patent applications regarding e-waste recycling from 148 per million to 787 per million. A majority of these patents have to do with cable processing, but there are a fair percentage relating to PCBs, solar panels, and lamps as these all contain critical raw materials [3].

However, there is a growing challenge regarding the proper collection and recycling of e-waste as electronics themselves become more diverse. Electronic devices cover a broad category of consumer goods, from e-cigarettes to e-vehicles, and there are also electronic devices that are "invisible", i.e. clothing items with heating functions and furniture with massage functions. These more invisible electronic devices pose an issue for current recycling practices due to their variable composition requiring different treatments when recycling. A device like a phone can all be handled as one piece during the recycling process, but invisible electronics require more separation to determine which parts are to be treated as e-waste or standard waste [3].

Why is E-Waste a Problem?

The improper disposal of e-waste has significant economic and environmental impacts that increase alongside the generation of e-waste. Globally, 62 million metric tons of e-waste were generated in 2024, and it is expected to grow to 82 million metric tons by 2030. The generation of waste is rising five times faster than improvements in recycling efforts, leading to both the wasting of valuable metals within electronics and allowing heavy metals that comprise electronics to seep into the environment [17].

Waste of Valuable Resources

Around 28 billion USD of metals from e-waste were turned into secondary raw materials globally in 2022, but the overall value of the total generated e-waste is an estimated 91 billion USD, leaving 63 billion USD wasted by the improper disposal of e-waste [3]. There are up to 69 elements from the periodic table found in electronic devices as seen in figure 2. Precious metals such as gold, silver, and copper, as well as critical raw materials like cobalt, bismuth, and palladium are all found in electronics [18].When e-waste is improperly disposed of, these valuable metals are wasted, yet they are still needed for the increasing electronics that enter the market. This puts a strain on the primary mining of metals for electronics as demand for these metals is only rising.

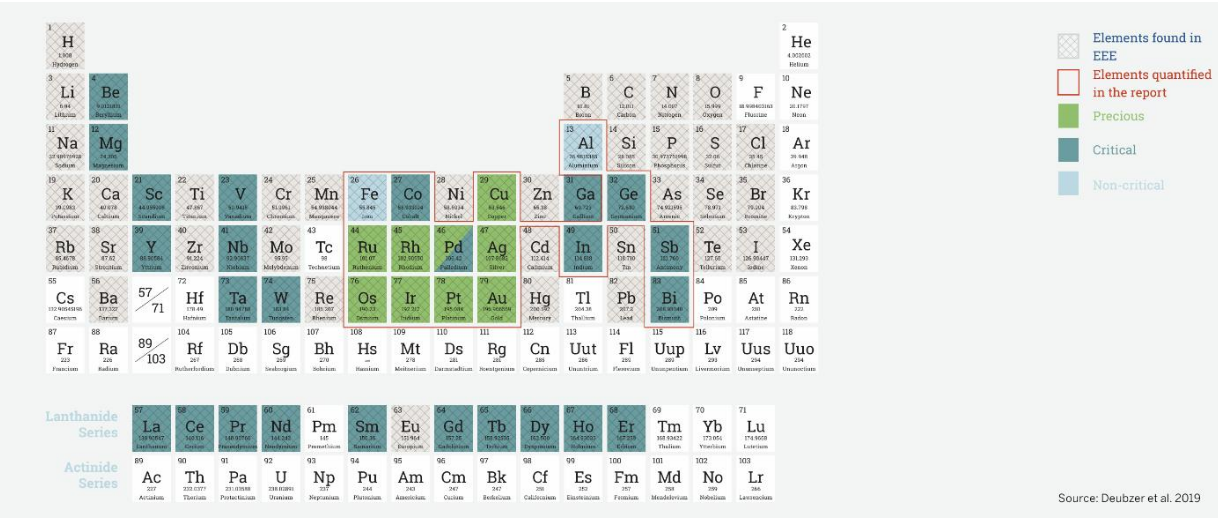

Figure 2*: Periodic table highlighting which metals are found in EEE, and the overlap between those precious, critical, and non-critical metals. Taken from [15].*

Environmental Effects

E-waste also has an ever-increasing negative effect on the environment. Improperly disposed of e-waste follows standard waste channels to landfills, where the environmental contaminates found in electronic devices, such as lead, mercury, polycyclic aromatic hydrocarbons (PAHs), and refrigerants, seep into the surrounding environment [15], [16]. These toxins bioaccumulate, meaning the toxins in the surrounding soil become concentrated in the organisms that interact with the environment, and if a toxin bioaccumulates, it can biomagnify or spread to higher levels of the food chain [16]. This is seen in places like Bangladesh where mercury, brominated flame-retardant plastics, and cadmium found in e-waste sites seeped into the soil contents and negatively affected the crop production. It is also seen in the paddy fields of Wenling, where high concentrations of PAHs were found in the paddy field soil near e-waste disposal sites [19]. In Qingyuan County, researchers found that chickens near e-waste sites had high levels of flame retardants in their bodies, and suggested that consumption of the chickens near the sites correlated to the flame retardant levels found in humans [16].

The toxins these electronic devices leech into the environment lead to increased cancer rates, birth defects and complications, and general decline in quality of life in areas near e-waste dump sites. Guiyu was a large site for informal recycling, and researchers found high rates of skin damage, headaches, vertigo, nausea, and ulcers in Guiyan workers. In a study from 2004 to 2008, Guiyan children were found to have blood chromium levels two to three times higher than in towns with no e-waste sites, which affects these children's growth and can damage their DNA [16]. Ultimately, increasing e-waste without the ability to properly dispose of the waste only serves to amplify the detrimental environmental effects it causes.

Mitigation Methods

As the rate of e-waste generation continues to increase, the environmental and economic impact of e-waste will grow in tandem. It is imperative that a greater effort is made by electronic manufacturers, consumers themselves, and policymakers to minimize e-waste.

When it comes to defining improvement of E-waste, figure 3 highlights the different ways e-waste is targeted, from bettering disposal practices to full prevention of e-waste. According to UNITAR's *Global E-Waste Monitor*, the ideal way to manage e-waste is to prevent it from being created, but this target is not always attainable, so there must be manners to prevent e-waste at all target points.

The following proposed mitigation methods attempt to cover the preferred manners of e-waste management. An implementation of a circular economy targets proper disposal, recovery, reuse, and recycling of e-waste. Increasing modular devices on the market targets prevention and reuse, and comprehensive legislation could be made to target all methods of managing e-waste.

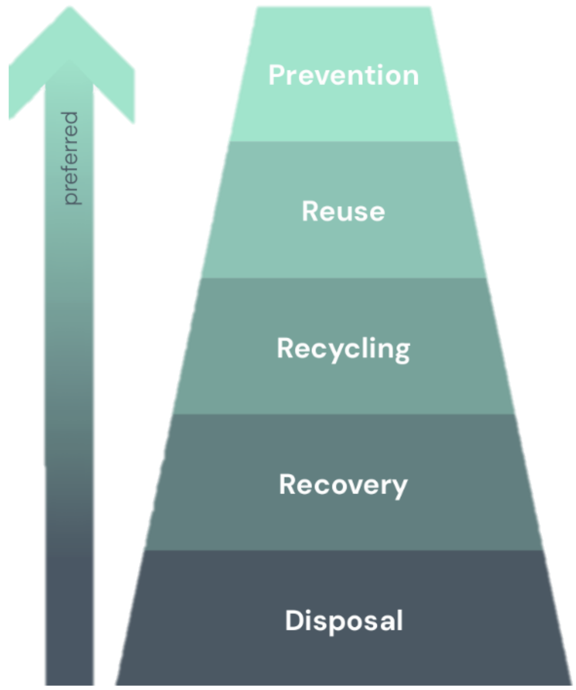

Figure 3*: The hierarchy of targeting e-waste, from targeting disposal to prevention, adopted from [3].*

Circular Economy

One suggested approach to minimizing the negative effects of e-waste is the implementation of a circular economy. A circular economy utilizes the idea that when a product is disposed of, useful materials can be recollected from the product. Thus, preventing the waste of metals like gold, copper, and silver.

A main point against the implementation of a circular economy is that the cost of recollecting the materials might outweigh the current disposal system of improperly dumping the majority of e-waste. If one factored the recorded values from 2022 of avoided greenhouse gas emissions, estimated to be 23 billion USD, and the 28 billion USD of recovered metals, and compared that to the 10 billion USD associated with treatment alongside the 78 billion USD in externalized costs to the population and environment, proper recycling of e-waste costs 37 billion USD worldwide [3]. However, considering that 63 billion USD was wasted since the estimated value of all global e-waste in 2022 was 91 billion USD, the cost of e-waste recycling could be offset by putting a greater effort on proper recollection of valuable materials.

Under the paradigm of a circular economy, e-waste would be a primary source of secondary raw materials. This should minimize the need for primary mining, and the issues of safety and morality that that poses. It would also minimize issues with market price fluctuations and material scarcity as the need for these materials increase with the increase of electronics being produced [18].

The circular economy itself creates incentive for proper collection of waste materials, but the responsibility of that collection is likely to fall on the producers. Many proposed circular economy models depend on the principle of extended producer responsibility (EPR). The EPR principle is essentially the idea that a producer of a device is still responsible for said device for the entire lifecycle of a product [3], [20]. In the case of a circular economy model, EPR would enforce the collection of e-waste to be done by the producers themselves. There are already producers and distributors of electronics that offer some sort of recollection service for their devices, though they are often small services that could be scaled if more economic focus shifted to the value of the metals wasted in improper e-waste disposal [13], [14].

Modular Technology

A large cause of the generation of e-waste is the disposal of devices due to a lack of upgrades, or a consumer desire to obtain newer products. A third of the world's e-waste, 20 billion kg, comes primarily from smaller electronics like e-cigarettes, vacuum cleaners, and toasters, and another 5 billion kg comes from devices like phones, laptops, and Wi-Fi routers [3]. To minimize the frequency of replacing devices, producers could make their technologies more modular and repairable.

According to authors Proske and Jager-Erben, a modular device should fulfill a certain design criteria such that it is made to benefit the user and ensure they will upgrade the device rather than replace it [5]. These design criteria are seen in figure 4, where it maps reasons for obsolescence to what in a modular design could serve to prevent the causes of obsolescence. The primary goal in the design of a modular device is to create something that is adaptable, upgradeable, expandable, and maintains functionality over time. These requirements can be implemented through modules on a device that can be easily exchanged and are inexpensive compared to the original device.

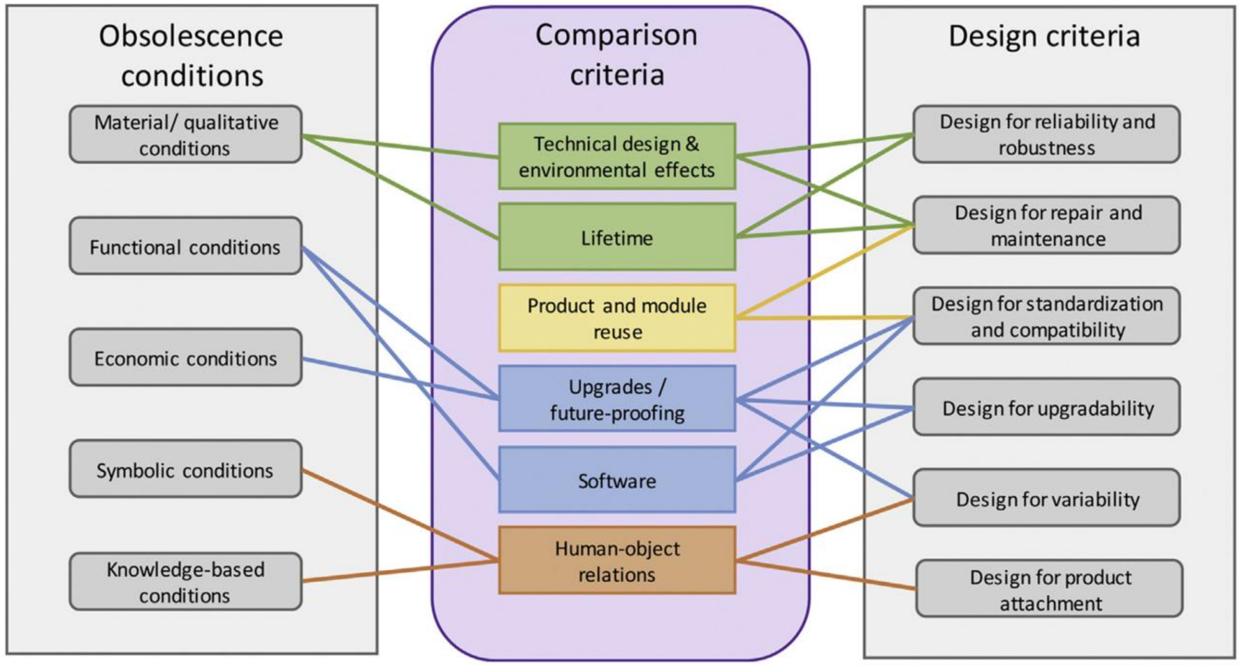

Figure 4*: The conditions that cause obsolescence linked to design criteria for a modular device that could prevent those causes, adopted from [5].*

There exist devices that are proof of concept on the market currently. The Fairphone is a phone brand started in 2013 with the intention of lowering the ethical and environmental impact of its devices, and they are on their 6th generation phone over a decade later, with a moderate customer base [21], [22]. The Framework laptop is a newer modular device, founded in 2020, that has been garnering increasing market attention due to the ease of repair and unique ability to exchange ports [23]. These devices show that it is possible to design a product that fits the modular design criteria and makes profit.

Comprehensive Legislation

Proper comprehensive legislation for e-waste could solve a majority of the issues in the U.S. regarding both waste generation and the proper disposal of e-waste. The need for legislation, however, leads to questions about what that legislation should hold, and who is responsible for upholding it?

Several groups have suggested policy regarding e-waste both globally and in U.S. ([1], [3], [11], [12], [18]). The following suggestions are a synthesis of these proposals of what e-waste legislation in the US could cover, and who the responsibility falls upon.

The first step to creating comprehensive legislation in the U.S. regarding e-waste is to properly define what e-waste is, as different entities define it in different manners. UNITAR's organization of e-waste in their *Global E-Waste Monitor* is concise, clear, and generally accepted as a manner of both defining e-waste and categorizing the different kinds. This definition should be adopted by policymakers, as it covers the umbrella term of e-waste and allows for subcategories if say a policy must cover telecommunication devices specifically but not large electronics like washing machines.

After clear definitions of what kinds of waste the legislation covers are established, then comes the matter of how the legislation enforces proper handling of e-waste. There needs to be some method of ensuring proper handling of waste by whatever entity is managing the e-waste. Enforcing EPR is a potential avenue for this. Original equipment manufacturers are likely the most equipped to handle the recycling of their own products, and they could design their products for better recyclability. Financially this could take a toll on producers, but the cost of recycling could be reworked into the original fee of a product. This cost should be visible to the consumer, so they are aware of the cost of recycling a product before buying it. This concept is called an advanced recovery fee, and it is implemented currently only in the state of California so scalability of this solution is understudied [1]. Third-party, municipal government, and non-profit organizations should also be allowed to collect e-waste under a proposed legislation, and there must be regulations ensuring that they both properly handle the waste in the U.S. and legally ship e-waste across the border with the consent of the country they desire to ship the waste to.

Additionally, for a consumer wishing to toss their electronic device, this must be a free and accessible process. National Institute of Standards and Technology engineer Kelsey Schumacher proposes that legislation should establish "standards that require the establishment of geographically convenient collection locations where covered entities can discard unwanted electronics free of charge. Provisions for collection in rural areas should be included," and that recycling should be free of charge to as many people as possible. There is a positive correlation between accessible recycling and collection rates [1].

If this is federal legislation, state environmental agencies should have the authority to adjust regulations to meet the demands of the industry as it changes. Electronic manufacturing is an ever-changing field, and the field of e-waste recycling should be able to change along with it.

Finally, in order for any amount of e-waste legislation to be effective, there must be an effort made to ensure that people know of what avenues to take for the disregarding of their waste. Either through governmental efforts or the recyclers themselves, there must be some manner of education on both the benefits of proper e-waste disposal and how to properly do it.

These suggestions could be implemented on a state, federal, or even third-party (such as through the EPA) level. They are all feasible, though they require effort from the parties of policymakers, electronic manufacturers, recycling stakeholders, and product consumers. Schumacher presents a table for who the responsibilities of what comprehensive legislation should fall upon that is seen in table 2. It aptly sorts which party is in charge of what, regarding physical, financial, and informative responsibilities.

	Physical Responsibility	**Financial Responsibility**	**Informative Responsibility**
Collection & Sorting	Municipal governments, private companies, and non-profit organizations	Manufacturers – individual or collective	Local and state/or state governments
Reuse, Recycling & Treatment	Collectors, refurbishers, recyclers	Manufacturers – individual or collective	Collectors and recyclers
Monitoring & Enforcement	Centralized management organization (i.e., state environmental agency)	Manufacturers – individual or collective	Local and state/or state governments

***Table 2**: The range of responsibilities for the various components in implementing comprehensive legislation regarding e-waste, as seen in [1].*

Conclusions

Improper disposal of e-waste is an issue that will only continue to grow if proper efforts are not made. This paper aims to draw attention to that issue by looking at current recycling practices, legislation, and the economic and environmental harm e-waste causes. Currently, there is an increasing amount of e-waste recycling technologies being patented and there does exist some legislation regarding e-waste, but in its current state that is not enough to mitigate the compounding waste of valuable metals and harmful chemicals that e-waste releases into the environment. The economic solution proposed is the implementation of a circular economy, which would allow the harvesting of valuable metals from electronic devices once they have been disposed of as to minimize both the waste of precious metals and the heavy metals that seep into the environment from improperly disposed of electronics. The manufacturing solution proposed is to design products to be modular, which would allow for the upgrading and fixing of specific parts of a device rather than the need to dispose of an entire device. Guidelines for comprehensive legislation are also proposed, highlighting the need to properly define e-waste and regulate the disposal of it.

The solutions proposed are by no means a concrete framework, so the next step from this paper would be to craft a framework from the mitigation methods that could be feasibly implemented. Small amounts of e-waste are already recycled and mined for their metals, legislation exists regarding e-waste recycling, and modular technologies exist in the market. There is proof of concept, but none of these solutions have been implemented on a large scale. Studying both how to implement scalable solutions, as well as why they have yet to be implemented is a logical next step from this paper.

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